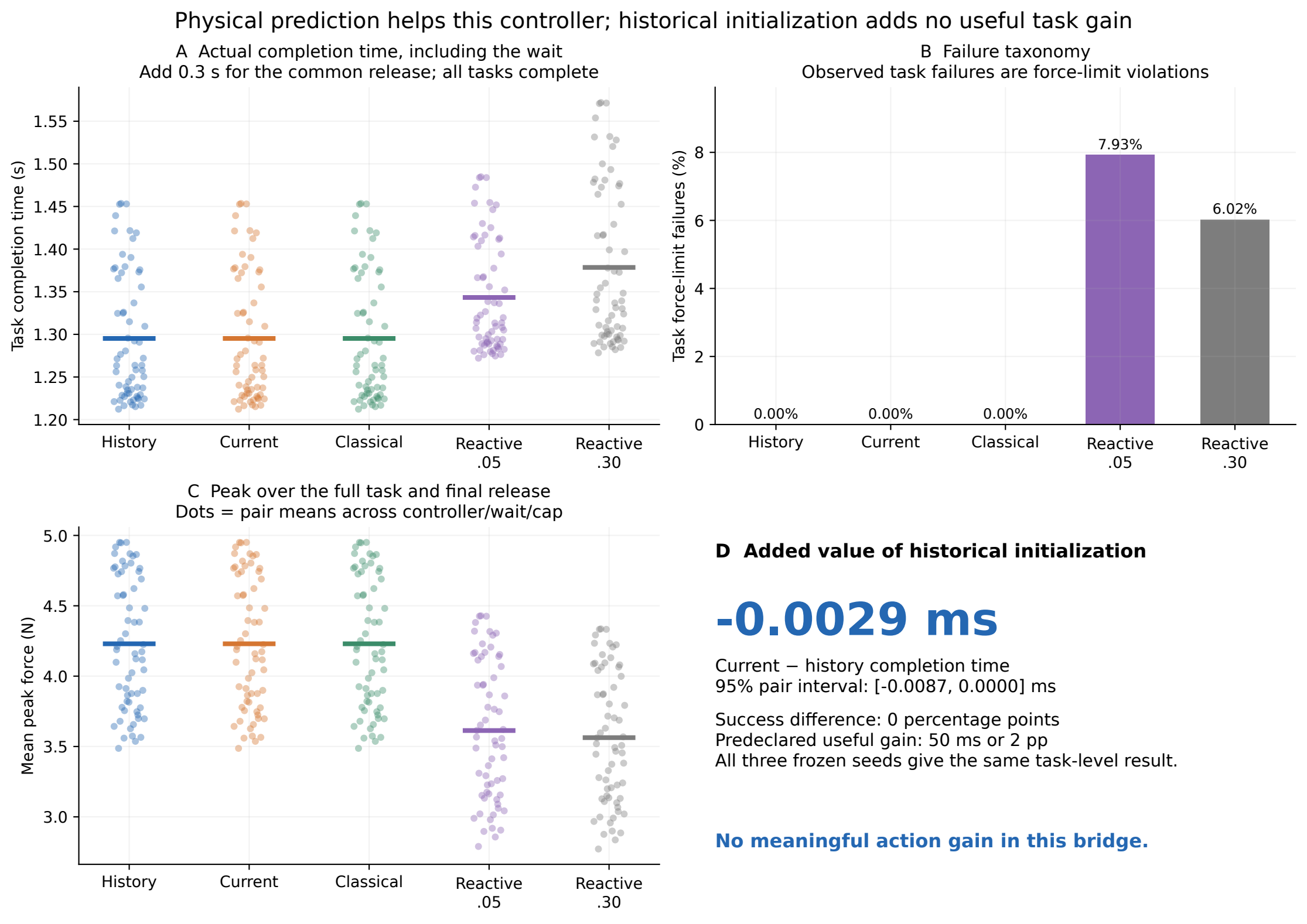

Physical prediction helps this controller; historical initialization adds no useful task gain
A Actual completion time, including the wait
Add 0.3 s for the common release; all tasks complete
Task completion time (s)
1.55
1.50
1.45
1.40
1.35
1.30
1.25
1.20
History
Current
Classical
Reactive .05
Reactive .30
B Failure taxonomy
Observed task failures are force-limit violations
Task force-limit failures (%)
8
6
4
2
0
0.00%
0.00%
0.00%
7.93%
6.02%
History
Current
Classical
Reactive .05
Reactive .30
C Peak over the full task and final release
Dots = pair means across controller/wait/cap
Mean peak force (N)
5.0
4.5
4.0
3.5
3.0
History
Current
Classical
Reactive .05
Reactive .30
D Added value of historical initialization
-0.0029 ms
Current − history completion time
95% pair interval: [-0.0087, 0.0000] ms
Success difference: 0 percentage points
Predeclared useful gain: 50 ms or 2 pp
All three frozen seeds give the same task-level result.
No meaningful action gain in this bridge.